

VOCABULARY AND FORMULAS APPENDIX

Genome Research — First-Pass In Silico Project, Frozen Record v1.0

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Purpose and scope

This appendix is a reader-facing reference for terminology, variables, and formulas used across Phases 1–6 of the project “Genomic and Molecular Analysis of a 17p13.3 Microdeletion and Its Candidate Visual-Neurological Mechanisms.” It preserves the frozen project’s terminology and inferential boundaries. Definitions describe how a term is used in this project, which may be narrower than a general textbook definition.

Formula classification rule

Formulas are labeled as structural calculations, statistical definitions, biochemical reactions, operational definitions, conceptual models, or Phase 5 modeling constructs. A conceptual equation is not a validated physiological law. An ordinal score is not a biological effect size or probability. Where the frozen record named a standard statistical method but did not preserve its explicit equation, this appendix supplies the canonical mathematical definition and labels it as an explanatory addition rather than silently attributing it to the original calculation pipeline.

Fixed structural reference

GRCh37/hg19 chr17:1,411,408–1,841,103; nominal span 429,696 bp; heterozygous deletion. The frozen locus set contains 13 protein-coding and 4 noncoding loci. INPP5K and RTN4RL1 are boundary-intersected. GRCh38 may be used for modern annotation only with documented build conversion and must not silently replace the frozen coordinates.

Appendix A — Vocabulary

This glossary contains 123 project terms. It emphasizes terms that affect interpretation, reproducibility, genotype-state equivalence, expression analysis, visual physiology, and causal claims.

Term	Project-specific description	Primary phase(s)
Adaptation	Adjustment of retinal or neural responsiveness during or after sustained or repeated stimulation. In this project, dark adaptation specifically means recovery of visual sensitivity after photobleaching.	Phase 4; retained in Phase 6
Allele	One of the copies or sequence states of a locus. For this CNV, interpretation generally concerns the affected allele versus the homologous chromosome carrying the remaining copy.	Phases 1, 4
Amino-acid-coding sequence	The CDS nucleotides that encode amino acids, excluding the stop codon when the project reports the INPP5K coding-loss percentage.	Phase 1
Annotation release	A defined version of a genome/gene annotation resource. The project records releases because gene boundaries, transcripts, and labels can change without the underlying CNV changing.	Phases 1, 3, 5
Autolysosomal reformation (ALR)	Membrane-recycling process that regenerates lysosomes from autolysosomes after prolonged autophagy. INPP5K is implicated in normal ALR.	Phases 2, 4
BED coordinate convention	A 0-based, half-open interval convention commonly used by BED/BEDTools. The project warns that GFF3 coordinates must be converted correctly before overlap calculations.	Phase 1
Biallelic loss	Disruption of both gene copies. It is not genotype-equivalent to the one-copy heterozygous loss produced by the index CNV.	Phases 4, 5
Biological replicate	An independently derived biological sample, donor, line, or clone that supports inference beyond technical repeat measurements. Many cells from one clone do not replace independent biological replication.	Phases 5, 6
Boundary-intersected gene	A gene crossed by a reported CNV boundary rather than fully encompassed by the interval. INPP5K and RTN4RL1 are the two boundary-intersected protein-coding loci in the frozen map.	Phase 1
Breakpoint	The location at which a structural variant transitions between retained and deleted sequence. The array-reported coordinates are operational boundaries, not nucleotide-exact junction sequences.	Phases 1, 6
Bulk RNA-seq	RNA sequencing performed on mixed populations of cells from a tissue. It measures aggregate expression but cannot by itself identify which constituent cell types produce the signal.	Phase 3
Candidate modifier	A factor plausibly capable of changing phenotype severity, threshold, penetrance, or expression without being established as a sufficient primary cause.	Phase 4

Term	Project-specific description	Primary phase(s)
Causal chain	The ordered evidentiary/biological sequence from genomic disruption through dosage, molecular perturbation, spatial opportunity, cellular physiology, systems physiology, and phenotype. Missing downstream links cannot be supplied by strong upstream evidence.	Phases 4, 5, 6
CCDS	Consensus Coding Sequence. In Phase 1 it is used as a standardized coding reference, including for the RTN4RL1 coding-overlap calculation.	Phase 1
Cell-type specificity	The degree to which expression or another molecular signal is concentrated in one cell class relative to others. It is evidence of biological opportunity, not evidence of dysfunction.	Phases 3, 5
Cellular phenotype	A measurable change in cell structure, survival, trafficking, signaling, electrophysiology, or stress response after perturbation.	Phase 4
CNV (copy-number variant)	A genomic deletion or duplication that alters the number of copies of one or more loci. The index CNV is a heterozygous deletion.	All phases
Coding sequence (CDS)	The nucleotide sequence of a transcript that encodes protein. The Phase 1 overlap calculations operate on CDS segments rather than total genomic gene span.	Phase 1
Coexpression	Correlated or overlapping expression of genes across samples, cells, or biological units. Coexpression does not establish molecular interaction, epistasis, synergy, or causality.	Phases 3, 5
Compensation	Biological buffering that reduces the functional impact of a genomic perturbation, including intact-allele expression, paralogs, developmental adaptation, or homeostatic responses.	Phases 4–6
Complete knockout	Experimental removal or inactivation of all functional copies of a gene. Knockout evidence is not treated as equivalent to heterozygous deletion evidence.	Phases 4, 5
Confidence interval / uncertainty interval	A quantitative interval representing uncertainty around an estimate. Phase 5 requires uncertainty to be retained where calculable rather than replaced by point estimates alone.	Phase 5
Constraint / dosage characteristic	Genomic or gene-level evidence relevant to tolerance of loss or copy-number change. In Phase 5, such characteristics may be used in matched null intervals when feasible.	Phase 5
Counterevidence	Evidence that contradicts or weakens a proposed mechanism, such as normal heterozygous phenotypes, compensation, reduced rather than increased physiological response, or protective effects under specific conditions.	Phases 4–6
Cryptographic hash	A deterministic digital fingerprint proposed for freezing Phase 5 inputs so later users can verify that the analyzed dataset has not silently changed.	Phase 5

Term	Project-specific description	Primary phase(s)
Dark adaptation	Recovery of visual sensitivity after photobleaching or sustained light exposure; treated as a possible retinal reserve endpoint.	Phases 4–6
Detection fraction	The fraction of cells/nuclei or samples in which a gene is detectably measured. In single-cell data it is influenced by both biology and technical dropout.	Phases 3, 5
Developmental trajectory	The pattern of a gene's expression or biological effect across developmental stages. Phase 3 specifies within-gene normalization to identify relative prenatal, neonatal, childhood, or adult peaks.	Phase 3
Directional convergence	Multiple perturbations pushing a shared endpoint in the same direction. This is stronger than merely affecting the same system, but still does not itself prove synergy.	Phase 4
dLGN / lateral geniculate nucleus (LGN)	The dorsal lateral geniculate nucleus, the principal thalamic relay between retinal ganglion cells and primary visual cortex.	Phases 3–6
Dominant-negative variant	A variant whose product interferes with the normal product. Its mechanism is not assumed to be equivalent to a simple heterozygous deletion.	Phases 4, 5
Dosage consequence	A demonstrated change in RNA, protein, molecular function, or cellular behavior attributable to copy-number/genotype state. Deletion alone does not establish the size of the dosage effect.	Phases 4–6
Dosage sensitivity	The degree to which changing gene copy number alters transcript, protein, molecular function, cell behavior, or phenotype.	Phases 1, 2, 4–6
Dropout	Failure to detect a transcript in single-cell/single-nucleus sequencing even when biologically present, especially for low-abundance transcripts. Zero counts in a small cell population are therefore not automatically biological absence.	Phase 3
Efferocytosis	Recognition and engulfment of apoptotic cells or debris by phagocytic cells; SCARF1 participates in C1q-associated efferocytosis.	Phases 2, 4
Electroretinography (ERG)	Measurement of retinal electrical responses to light. The a-wave primarily reflects photoreceptor activity; the b-wave reflects downstream inner-retinal activity.	Phases 4, 6
Endolysosomal system	The connected endosome-to-lysosome network that sorts, transports, degrades, and recycles cellular cargo.	Phases 2, 4–6
Enrichment	A signal occurring more strongly or frequently than a defined comparison or null expectation. Phase 5 requires matched null analysis before treating convergence as unusually specific.	Phases 3, 5
Entrapment of missingness / missing $\neq$ zero	Project rule that an unmeasured causal layer must remain missing/unresolved rather than being encoded as zero or inferred from neighboring layers. Zero is reserved for defined null or evidence-of-absence states.	Phase 5

Term	Project-specific description	Primary phase(s)
Epileptogenesis	The biological process by which a network develops a persistent tendency to generate seizures. MIR22 evidence in the project is most directly related to post-insult epileptogenesis rather than photic seizure causation.	Phases 4–6
Epileptogenic reserve	Conceptual resistance of a neural network to transitions into persistent or recurrent epileptic activity; lower reserve implies easier pathological transition under challenge.	Phases 4–6
Epistasis	A genetic interaction in which the effect of one locus depends on the state of another. The project did not demonstrate epistasis among deleted loci.	Phases 2, 4–6
Evidence hierarchy	The ordered project rule: genomic evidence → molecular mechanism → expression/spatiotemporal opportunity → perturbational evidence → cellular phenotype → developmental consequence → visual-system physiology → phenotype association → causality.	Phases 4–6
Expression prevalence	The proportion of relevant biological units showing detectable expression; closely related to detection fraction and distinct from expression magnitude.	Phase 5
False discovery rate (FDR)	Expected proportion of false discoveries among findings declared significant. Phase 3 specifies Benjamini-Hochberg control with $q < 0.05$ for planned coexpression testing.	Phase 3
Functional convergence	Independent genes or mechanisms affecting the same molecular system, cellular process, anatomical compartment, developmental operation, or phenotype endpoint.	Phases 2–6
Functional reserve	Difference between a system's available/maximal biological capacity and the physiological demand placed on that system. The project's leading model proposes distributed reduction or alteration of reserve rather than obligatory baseline failure.	Phases 4–6
Gain	Ratio between change in output and change in input. Increased sensory gain means a larger physiological response to the same matched stimulus.	Phase 4
Gain-of-function variant	A variant that increases or changes activity in a way not equivalent to simple loss of one copy.	Phases 4, 5
Genome assembly	A defined coordinate reference for the genome. The frozen structural reference is GRCh37/hg19; GRCh38 may be used for modern annotation only with documented conversion.	All phases
Genotype-state discipline	Project rule requiring explicit distinction among heterozygous deletion, partial deletion, hypomorph, biallelic loss, complete knockout, dominant-negative/gain-of-function variants, and pharmacologic perturbation.	Phases 4–6

Term	Project-specific description	Primary phase(s)
GFF3 coordinate convention	A 1-based, inclusive annotation coordinate convention used in the Phase 1 computational overlap workflow.	Phase 1
Haploinsufficiency	A state in which one functional gene copy is insufficient for normal molecular, cellular, or clinical function. It must be demonstrated rather than presumed from deletion.	Phases 2, 4–6
Heterozygous deletion / heterozygous loss	Loss or disruption of one of two gene copies. It does not automatically imply exactly 50% RNA, protein, or function.	All phases
Hypomorph	An allele retaining partial gene function rather than complete loss.	Phases 4, 5
In silico	Analysis conducted computationally using genomic, transcriptomic, literature-derived, or simulated data rather than direct wet-lab perturbation of the exact CNV.	Phases 5, 6
Independent replication	Reproduction of an observation in independent donors, lines, datasets, studies, or experiments rather than repeated measurements of the same underlying sample.	Phases 3, 5, 6
Isogenic	Genetically matched experimental lines differing only at the engineered locus/CNV of interest, minimizing background-genome confounding.	Phases 4, 6
Koniocellular (K), magnocellular (M), parvocellular (P)	Major subdivisions of primate visual-thalamic organization. Phase 3 did not identify the affected loci as a preferential adult human magnocellular transcriptional program.	Phases 3, 4
Leave-one-locus-out analysis	Sensitivity procedure in which each affected locus is removed in turn and the model is recomputed to determine whether conclusions are distributed or driven by one/small set of loci.	Phase 5
Leave-one-source-out analysis	Sensitivity procedure that removes an evidence source/study and recomputes conclusions to test dependence on individual publications or datasets.	Phase 5
lncRNA (long noncoding RNA)	A longer non-protein-coding RNA transcript. PITPNA-AS1 and MIR22HG are treated as potentially detectable but often low-abundance or assay-sensitive.	Phases 1, 3
Locus	A defined genomic location. The frozen deletion affects at least 17 annotated loci.	All phases
MANE Select	Matched Annotation from NCBI and EMBL-EBI representative transcript used as the preferred standardized transcript reference when available.	Phase 1
Marker gene	A gene reproducibly enriched in or characteristic of a cell class/subtype. Failure of an affected locus to appear in a marker list is not the same as proof of absence.	Phase 3
Mechanistic domain	A project-defined biological subsystem into which locus-level evidence can be grouped, such as retinal reserve, membrane/endosomal reserve, damage-response reserve, or neural/network reserve.	Phase 5

Term	Project-specific description	Primary phase(s)
Mechanistically plausible	Supported by a coherent chain of biological evidence but not established as causal for the final phenotype.	Phases 4–6
MicroRNA (miRNA)	A small regulatory RNA. MIR22 is within the deletion; conventional poly(A) RNA-seq is not a valid direct measurement of mature miR-22 abundance.	Phases 1, 3–6
Monte Carlo simulation	Repeated stochastic sampling used in Phase 5 to examine methodological behavior such as power, false-positive behavior, and model identification. It did not generate biological evidence for the CNV.	Phase 5
Multiple-testing correction	Adjustment for testing many hypotheses/frequencies so nominally significant findings are not overcounted. Phase 3 specifies BH-FDR for coexpression; Phase 6 requires correction for future frequency scans.	Phases 3, 5, 6
Neural/network reserve	Operational Phase 5 domain concerning neurite development, synapse density, network organization, calcium dynamics, and white-matter/conduction-related evidence.	Phase 5
Neuroprotectin D1 (NPD1)	A DHA-derived lipid mediator with anti-inflammatory and anti-apoptotic actions in retinal pigment epithelium under oxidative stress.	Phases 2, 4
Normalized expression	Expression transformed or scaled according to a defined pipeline to permit within-dataset comparison. Values from unrelated sequencing platforms should not be compared as if directly interchangeable.	Phases 3, 5
nTPM	Normalized transcripts per million as reported by Human Protein Atlas resources used in Phase 3 screening. It is a source-specific normalized expression measure and not an absolute molecule count.	Phase 3
Null / weak-specificity model	Competing model in which the deleted genes have real biology but the apparent visual-neurological convergence is not specific evidence for the index phenotype.	Phases 5, 6
Null result	A measured result that does not support the tested effect under specified conditions. It is retained as evidence rather than discarded or redefined.	All phases
Operational CNV boundary	The reported array-coordinate edge used for calculations. It is not asserted to be the nucleotide-exact physical chromosome junction.	Phases 1, 6
Outer segment	Specialized photoreceptor compartment containing photopigment-rich membranes where photon absorption and phototransduction begin.	Phases 4, 6
Pairwise coexpression	Coexpression evaluated between two genes, typically across independent biological units. It is distinct from multi-gene coexpression.	Phase 5
Paralog compensation	Functional buffering by a related gene with overlapping function. It is a potential reason why severe knockout phenotypes may not appear after one-copy loss.	Phases 4–6

Term	Project-specific description	Primary phase(s)
Parameter sensitivity	Dependence of a model conclusion on defensible changes to weights, mappings, penalties, or missing-data rules. A conclusion is called robust only if it survives reasonable variation.	Phase 5
Partial gene deletion	A deletion affecting only part of a gene/locus. The project treats INPP5K and RTN4RL1 as partial genomic deletions with substantial coding disruption on the affected allele.	Phases 1, 5
PEDE	Pigment epithelium-derived factor, encoded by SERPINF1; a secreted retinal trophic, antiangiogenic, lipid-linked, and cytoprotective protein.	Phases 2, 4–6
Perturbation equivalence	The degree to which an experimental perturbation matches the index CNV genotype state. Phase 5 requires separate weighting/mapping rather than treating knockout and heterozygous deletion as equivalent.	Phase 5
Phase locking	Consistency of response phase relative to a periodic stimulus or oscillation; proposed as an endpoint for future flicker/network experiments.	Phases 4, 5, 6
Phenotype association	Observed statistical or clinical relationship between genotype and phenotype. Association is downstream of molecular evidence but is not automatically causation.	Phases 4–6
Phosphoinositide	A phosphorylated phosphatidylinositol lipid involved in membrane identity, trafficking, signaling, and protein recruitment.	Phases 2, 4–6
Photochemical vulnerability	Greater molecular or cellular injury from the same photon/light exposure, distinct from an abnormally large sensory response.	Phase 4
Photoparoxysmal response (PPR)	An abnormal epileptiform EEG response evoked by intermittent photic or pattern stimulation; it may occur with or without a clinical seizure.	Phases 4–6
Photosensitive epilepsy (PSE)	Epilepsy in which visual stimulation can provoke abnormal EEG activity and, in some patients, clinical seizures.	Phases 4–6
Phototransduction	Conversion of absorbed photons into an electrical photoreceptor response through opsin/rhodopsin, transducin, PDE6, cGMP, and cyclic-nucleotide-gated-channel signaling.	Phase 4
PI(3,4,5)P3 / PIP3	Phosphatidylinositol 3,4,5-trisphosphate, another substrate represented in the INPP5K reaction set.	Phase 4
PI(4,5)P2	Phosphatidylinositol 4,5-bisphosphate, a membrane signaling lipid whose 5-phosphate can be removed by INPP5K.	Phases 2, 4
PI3P	Phosphatidylinositol 3-phosphate, enriched on early endosomes; the WDR81/WDR91-associated system regulates endosomal PI3P.	Phases 2, 4
PI4P	Phosphatidylinositol 4-phosphate, a signaling lipid produced by INPP5K-mediated dephosphorylation of PI(4,5)P2.	Phases 2, 4

Term	Project-specific description	Primary phase(s)
Primary visual cortex (V1)	Principal cortical recipient of LGN visual input and a major node relevant to visually driven cortical responses and photoparoxysmal propagation.	Phases 3–6
Pseudobulk	Aggregation of single-cell observations into biologically meaningful units such as donor × cell type × region before gene-level statistical analysis, reducing pseudoreplication.	Phase 3
Pseudogene	A gene-like sequence generally lacking canonical protein-coding function. RN7SL105P is retained as an affected pseudogene but requires mapping caution.	Phases 1, 3
Pseudoreplication	Treating non-independent observations (for example many cells from one clone) as independent biological replicates, artificially inflating apparent precision.	Phases 3, 5, 6
q-value	FDR-adjusted significance measure. Phase 3 specifies $q < 0.05$ as the primary criterion for planned coexpression analyses where sample size permits.	Phase 3
Random-interval null model	Phase 5 control strategy comparing the index CNV with matched genomic intervals processed through the same convergence-scoring pipeline.	Phase 5
Rescue	Correction or restoration of a gene/CNV state or pathway that reverses a phenotype. Rescue provides stronger causal specificity than association alone.	Phases 4–6
Retinal ganglion cell (RGC)	Output neuron of the retina whose axons form the optic nerve and project to visual brain targets.	Phases 3, 4
Retinal pigment epithelium (RPE)	Support epithelium adjacent to photoreceptor outer segments; critical for phagocytosis, retinoid cycling, lipid handling, trophic support, and metabolic homeostasis.	Phases 2–6
Retinal/photoreceptor reserve	Operational Phase 5 domain including photoreceptor survival, outer-segment integrity, ERG amplitude, dark adaptation, recovery after light challenge, and oxidative-stress response.	Phase 5
Robust conclusion	In Phase 5 terminology, a conclusion that remains stable across reasonable parameterizations/sensitivity analyses.	Phase 5
Sensory hypersensitivity	An abnormally large perceptual or physiological response to the same light stimulus. This is distinct from photochemical vulnerability.	Phase 4
Sequential-process convergence	Different genes acting at distinct stages of the same biological process. Phase 5 distinguishes this from direct physical interaction or synergy.	Phase 5
Single-cell RNA-seq (scRNA-seq)	RNA sequencing of individual cells. It provides cell-type localization but is subject to dropout, donor non-independence, batch effects, and assay-specific capture biases.	Phase 3
Single-nucleus RNA-seq (snRNA-seq)	RNA sequencing of isolated nuclei. Useful for human brain tissue but may differ from whole-cell RNA-seq in transcript capture and abundance patterns.	Phase 3

Term	Project-specific description	Primary phase(s)
Spatial co-localization	Presence of relevant genes/mechanisms in the same tissue or cell type, providing opportunity for interaction but not proving it.	Phases 3, 5
Spatiotemporal opportunity	Evidence that a gene/mechanism is expressed in the relevant tissue/cell type and developmental or physiological window. Expression creates opportunity, not dysfunction.	Phases 3–6
Spearman rank correlation	Nonparametric rank-based measure specified in Phase 3 for coexpression when sample size permits, avoiding an assumption of linear relationships.	Phase 3
Spliceosome	Large ribonucleoprotein machinery that removes introns from pre-mRNA; PRPF8 is a central U5 spliceosomal scaffold.	Phases 2, 4
Steady-state visual evoked potential (SSVEP)	Stimulus-frequency-locked cortical response produced by periodic visual stimulation; relevant to entrainment and temporal-stability testing.	Phase 4
Stress-dependent vulnerability	A phenotype that may be weak/absent at baseline but emerges under oxidative stress, injury, repeated activity, aging, metabolic demand, or other challenge.	Phases 4–6
Synergy	A combined effect greater than expected under a specified additive model. The project does not claim demonstrated synergy among deleted loci.	Phases 2, 4–6
Temporal functional convergence	Different loci affecting the same endpoint at different developmental or physiological stages rather than acting simultaneously.	Phase 4
Topological convergence	Different perturbations affecting the same biological network or structural endpoint even if they push that endpoint in different directions.	Phase 4
TPM (transcripts per million)	RNA-seq abundance normalization used in Phase 3 tissue-expression resources. Cross-platform absolute TPM values are not assumed directly comparable.	Phase 3
Visual-neural reserve	Integrated capacity of retinal, visual-pathway, and cortical systems to maintain stable function under physiological or pathological demand.	Phases 4–6
Wet-lab validation	Direct experimental testing in engineered/biological systems. Phase 6 proposes exact-CNV isogenic systems with rescue/correction rather than inferring exact-genotype physiology from literature alone.	Phase 6
Within-gene z-score	A standardized value proposed in Phase 3 to show whether a gene is relatively high or low compared with its own expression across developmental stages, not to compare absolute expression between different genes.	Phase 3

Appendix B — Formula and Calculation Register

This register contains 45 formulas, reactions, quantitative definitions, and model expressions. Each entry states what the expression means and whether it was an actual project calculation, a specified analysis method, or a conceptual/future-test model.

B.1 Nominal interval span

$$L = D_e - D_s + 1$$

Variables / definitions: D_s = reported inclusive start coordinate; D_e = reported inclusive end coordinate.

Interpretation: For chr17:1,411,408–1,841,103, this yields 429,696 bp.

Status / limitation: Structural calculation; exact for the reported inclusive coordinates, not the physical junction length.

Project provenance: Phase 1

B.2 Per-CDS-segment overlap

$$L_overlap = \max(0, \min(C_e, D_e) - \max(C_s, D_s) + 1)$$

Variables / definitions: C_s/C_e = inclusive CDS-segment start/end; D_s/D_e = inclusive deletion start/end.

Interpretation: Counts the number of deleted bases within one CDS segment under 1-based inclusive coordinates.

Status / limitation: Structural calculation used by the replication-ready Phase 1 method.

Project provenance: Phase 1

B.3 Total deleted coding length

$$L_deleted = \sum_i L_overlap,i$$

Variables / definitions: i indexes CDS segments.

Interpretation: Sums deleted bases across all CDS segments of the reference transcript.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.4 Total CDS length

$$L_CDS = \sum_i (C_e,i - C_s,i + 1)$$

Variables / definitions: i indexes CDS segments.

Interpretation: Total coding length of the transcript under inclusive coordinates.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.5 Percent CDS deleted

$$\%CDS\ deleted = (L_deleted / L_CDS) \times 100$$

Variables / definitions: $L_deleted$ = deleted coding bases; L_CDS = total coding bases.

Interpretation: Quantifies the fraction of coding sequence intersected by the CNV. Stop-codon handling must be stated.

Status / limitation: Structural calculation.

Project provenance: Phase 1

B.6 INPP5K amino-acid coding denominator

$448 \text{ aa} \times 3 \text{ nt/aa} = 1,344 \text{ coding nt}$

Variables / definitions: 448 = protein length used in Phase 1; stop codon excluded for amino-acid-coding calculation.

Interpretation: Provides the denominator for the INPP5K amino-acid-coding loss estimate.

Status / limitation: Worked structural calculation.

Project provenance: Phase 1

B.7 INPP5K coding-loss estimate

$666 / 1,344 = 0.4955 \approx 49.6\%$

Variables / definitions: 666 = approximate deleted coding nucleotides under the reported-coordinate model.

Interpretation: Approximate fraction of MANE amino-acid-coding sequence lying within the deleted interval on the affected allele.

Status / limitation: Worked structural estimate; dependent on reported CNV boundaries and transcript model.

Project provenance: Phase 1

B.8 RTN4RL1 CCDS-loss estimate

$1,313 / 1,326 \times 100 \approx 99.0\%$

Variables / definitions: 1,313 = approximate CCDS nucleotides on deleted side; 1,326 = total CCDS nucleotide count stated in Phase 1.

Interpretation: Shows that small genomic-span intersection can correspond to near-total coding disruption.

Status / limitation: Worked structural estimate; coordinate-model dependent.

Project provenance: Phase 1

B.9 BED conversion for an inclusive deletion interval

$\text{BED_start} = D_s - 1; \text{BED_end} = D_e$

Variables / definitions: D_s/D_e are 1-based inclusive coordinates.

Interpretation: Converts the deletion to 0-based half-open BED form; Phase 1 gives chr17 1411407 1841103.

Status / limitation: Coordinate-system conversion rule.

Project provenance: Phase 1

B.10 Developmental within-gene z-score (canonical definition)

$z_{g,t} = (x_{g,t} - \text{mean}_g) / \text{SD}_g$

Variables / definitions: $x_{g,t}$ = expression for gene g at stage t ; mean_g and SD_g are calculated across the stages included for that gene.

Interpretation: Implements the Phase 3 concept of expressing each developmental value relative to that gene's own trajectory.

Status / limitation: Canonical mathematical definition supplied here for reader clarity; the frozen Phase 3 text specifies a within-gene z-score but does not preserve an explicit equation or exact preprocessing pipeline.

Project provenance: Phase 3

B.11 Detection fraction

$p_{\text{detect}} = n_{\text{detected}} / n_{\text{total}}$

Variables / definitions: n_{detected} = cells/nuclei or biological units passing the defined detection threshold; n_{total} = units tested.

Interpretation: Quantifies expression prevalence. Threshold and unit of independence must be declared.

Status / limitation: Canonical implementation of the Phase 3/5 metric; threshold was dataset-dependent.

Project provenance: Phases 3, 5

B.12 Spearman rank correlation

$\rho_s = \text{corr}(\text{rank}(X), \text{rank}(Y))$

Variables / definitions: X, Y = expression values across independent biological units.

Interpretation: Rank-based coexpression measure specified because linearity is not assumed.

Status / limitation: Statistical method specified in Phase 3; exact handling of ties/missing data must follow the software implementation.

Project provenance: Phase 3

B.13 Benjamini-Hochberg FDR adjustment

Sort $p_{(1)} \leq \dots \leq p_{(m)}$; compare $p_{(i)}$ to $(i/m)Q$, or compute adjusted q-values monotonically from $m \cdot p_{(i)}/i$

Variables / definitions: m = number of tests; i = ordered test rank; Q = target FDR (Phase 3 primary criterion $q < 0.05$).

Interpretation: Controls expected false-discovery proportion across multiple coexpression tests.

Status / limitation: Canonical algorithmic definition supplied for reader clarity; project specifies BH-FDR, not a custom implementation.

Project provenance: Phase 3

B.14 INPP5K 5-phosphatase reaction

$\text{PI}(4,5)\text{P}_2 + \text{H}_2\text{O} \rightarrow \text{PI}_4\text{P} + \text{P}_i$

Variables / definitions: P_i = inorganic phosphate.

Interpretation: Represents established INPP5K 5-phosphatase chemistry.

Status / limitation: Biochemical reaction; not a quantitative in vivo flux model.

Project provenance: Phase 4

B.15 INPP5K PIP3 reaction

$\text{PI}(3,4,5)\text{P}_3 + \text{H}_2\text{O} \rightarrow \text{PI}(3,4)\text{P}_2 + \text{P}_i$

Variables / definitions: Substrate/product phosphoinositides; P_i = inorganic phosphate.

Interpretation: Represents an established substrate reaction of INPP5K.

Status / limitation: Biochemical reaction; do not infer in vivo flux from kinetic constants alone.

Project provenance: Phase 4

B.16 VPS34 reaction

$\text{PI} + \text{ATP} \rightarrow \text{PI}_3\text{P} + \text{ADP}$

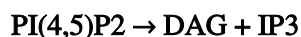
Variables / definitions: PI = phosphatidylinositol.

Interpretation: Represents kinase chemistry associated with endosomal PI_3P production.

Status / limitation: Biochemical reaction.

Project provenance: Phase 4

B.17 PLC branch



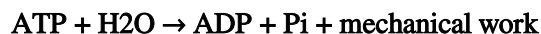
Variables / definitions: DAG = diacylglycerol; IP3 = inositol trisphosphate.

Interpretation: Canonical signaling branch relevant to calcium-regulatory context.

Status / limitation: Biochemical pathway representation.

Project provenance: Phase 4

B.18 Dynein/ATPase work



Variables / definitions: P_i = inorganic phosphate.

Interpretation: Represents ATP-powered motor/remodeling work; RILP recruits dynein but does not catalyze ATP hydrolysis itself.

Status / limitation: Biochemical/energetic representation.

Project provenance: Phase 4

B.19 PEDF-R/PNPLA2 phospholipase representation



Variables / definitions: Generic phospholipid substrate/products.

Interpretation: Simplified representation of phospholipase/lipase chemistry.

Status / limitation: Biochemical simplification, not a complete enzyme mechanism.

Project provenance: Phase 4

B.20 RPA binding equilibrium



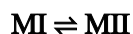
Variables / definitions: ssDNA = single-stranded DNA.

Interpretation: Binding equilibrium underlying RPA-mediated ATR recruitment and DNA-damage signaling.

Status / limitation: Conceptual biochemical equilibrium.

Project provenance: Phase 4

B.21 Rhodopsin conformational equilibrium



Variables / definitions: MI/MII = metarhodopsin I/II states.

Interpretation: Simplified photopigment conformational equilibrium influenced by membrane environment.

Status / limitation: Biochemical/biophysical representation.

Project provenance: Phase 4

B.22 Lipid-radical initiation



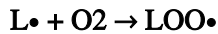
Variables / definitions: LH = lipid; L• = lipid radical.

Interpretation: Simplified initiation step in lipid peroxidation.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.23 Lipid-peroxyl propagation



Variables / definitions: $L\bullet$ = lipid radical; $LOO\bullet$ = lipid peroxyl radical.

Interpretation: Simplified oxygen-addition step in lipid peroxidation.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.24 Methionine activation



Variables / definitions: Met = methionine; SAM = S-adenosylmethionine.

Interpretation: Simplified activation route generating methyl donor SAM.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.25 SAM-dependent lysine methylation



Variables / definitions: SAH = S-adenosylhomocysteine.

Interpretation: Simplified methyltransferase reaction relevant to SMYD4.

Status / limitation: Biochemical representation.

Project provenance: Phase 4

B.26 Steady-state flux balance

$$dX/dt = J_{in} - J_{out}$$

Variables / definitions: X = pool/concentration; J_{in}/J_{out} = input/output fluxes.

Interpretation: At steady state $dX/dt = 0$ even when turnover is high; normal concentration does not prove normal flux.

Status / limitation: Quantitative definition/conceptual mass balance.

Project provenance: Phase 4

B.27 Functional-reserve condition

phenotype risk rises when $D(t) > C_{max}$

Variables / definitions: $D(t)$ = physiological demand at time t; C_{max} = available system capacity.

Interpretation: Formalizes the reserve concept: phenotype may emerge when demand exceeds capacity.

Status / limitation: Conceptual model; not a calibrated probability law.

Project provenance: Phase 4

B.28 Light-injury function

$I_{CNV}(H) > I_{WT}(H)$ at matched H indicates greater vulnerability

Variables / definitions: $I(H)$ = measured injury; H = light/photon exposure.

Interpretation: A direct vulnerability phenotype requires greater injury in CNV versus WT at the same exposure.

Status / limitation: Operational definition for future testing.

Project provenance: Phase 4

B.29 Light-injury threshold

H_{crit} = exposure producing a predefined injury criterion; vulnerability if H_{crit,CNV} < H_{crit,WT}

Variables / definitions: H_{crit} = challenge dose at a prespecified injury endpoint.

Interpretation: Lower threshold in CNV would support greater photochemical vulnerability.

Status / limitation: Operational definition.

Project provenance: Phase 4

B.30 Neural response gain

G = $\Delta\text{output} / \Delta\text{input}$

Variables / definitions: Input/output must be explicitly defined and measured under matched conditions.

Interpretation: Sensory/neural amplification requires G_{CNV} > G_{WT} for the same input domain.

Status / limitation: Operational quantitative definition.

Project provenance: Phase 4

B.31 Retinal temporal gain

G_R(f) = $|R(f)| / |S(f)|$

Variables / definitions: R(f) = retinal output amplitude at frequency f; S(f) = stimulus amplitude at f.

Interpretation: Quantifies frequency-dependent retinal transfer gain.

Status / limitation: Operational quantitative definition for future flicker testing.

Project provenance: Phase 4

B.32 Serial bottleneck capacity

J_{max} $\approx \min(C_1, C_2, \dots, C_n)$

Variables / definitions: C_i = capacity of sequential processing stage i.

Interpretation: Represents a serial process whose throughput is limited by the weakest stage.

Status / limitation: Conceptual model; approximation.

Project provenance: Phase 4

B.33 Layered lesion burden

dL/dt = D - P - R

Variables / definitions: L = lesion/damage burden; D = damage generation; P = prevention/tolerance; R = repair capacity.

Interpretation: Qualitative reserve representation balancing damage and protective/reparative processes.

Status / limitation: Conceptual model; not a validated physiological differential equation.

Project provenance: Phase 4

B.34 Factorial interaction model

Y = $\beta_0 + \beta_A \cdot A + \beta_B \cdot B + \beta_{AB} \cdot (A \times B) + \varepsilon$

Variables / definitions: Y = outcome; A/B = perturbation indicators or doses; β_{AB} = interaction coefficient; ϵ = residual.

Interpretation: β_{AB} tests departure from the specified additive model; useful for testing non-additive interaction/synergy claims.

Status / limitation: Standard statistical model proposed for reproducible interaction testing.

Project provenance: Phase 4

B.35 Developmental timing model

$$P = f(D_{\text{early}}, D_{\text{late}}, D_{\text{adult}})$$

Variables / definitions: P = mature phenotype; D terms = dosage effects during developmental windows.

Interpretation: Decomposes mature phenotype conceptually into early, late, and adult contributions.

Status / limitation: Conceptual model.

Project provenance: Phase 4

B.36 Photosensitive-network threshold

$$G_N \times I > \Theta_E$$

Variables / definitions: I = visual input; G_N = network propagation gain; Θ_E = threshold for pathological synchronization.

Interpretation: Conceptual condition under which visually driven network activity could cross a pathological threshold.

Status / limitation: Conceptual model; not calibrated to the index CNV.

Project provenance: Phase 4

B.37 Locus-level evidence function

$$E_{i,k} = f(D_i, M_i, S_i, P_i, V_i, N_i, R_i, C_i)$$

Variables / definitions: D_i = dosage evidence; M_i = molecular perturbation; S_i = spatiotemporal opportunity; P_i = perturbation equivalence; V_i = visual relevance; N_i = neural/network relevance; R_i = rescue/causal specificity; C_i = contradiction penalty.

Interpretation: Defines support from locus i to mechanistic domain k without assuming simple additivity.

Status / limitation: Phase 5 modeling framework; f was intentionally not fixed to a biologically unjustified numerical law.

Project provenance: Phase 5

B.38 Domain-level evidence aggregation

$$F_k = g(E_{1,k}, E_{2,k}, \dots, E_{n,k})$$

Variables / definitions: $E_{i,k}$ = locus-level contributions to domain k.

Interpretation: Aggregates locus evidence while requiring attention to correlated evidence and double-counting.

Status / limitation: Phase 5 modeling framework; g may be weighted, gated, minimum-link, geometric, or other justified form.

Project provenance: Phase 5

B.39 Causal-layer sequence

$$G \rightarrow D \rightarrow M \rightarrow S \rightarrow F_R \rightarrow F_N \rightarrow F_C \rightarrow F_P$$

Variables / definitions: G = genomic disruption; D = dosage; M = molecular/cellular perturbation; S = spatiotemporal opportunity; F_R = retinal consequence; F_N = neural/circuit consequence; F_C = cortical/network transformation; F_P = photic phenotype relevance.

Interpretation: Represents evidence propagation through necessary biological layers.

Status / limitation: Phase 5 causal architecture.

Project provenance: Phase 5

B.40 Bottleneck score example

$B = \min(G, D, M, S, F_R, F_N, F_C, F_P)$

Variables / definitions: Terms are causal-layer support measures on a common defined scale.

Interpretation: Prevents strong early-layer evidence from compensating numerically for an absent necessary downstream layer.

Status / limitation: Candidate Phase 5 formulation; explicitly not to be adopted automatically.

Project provenance: Phase 5

B.41 Convergence matrix

K_{ij}

Variables / definitions: K_{ij} records evidence that loci i and j converge biologically and the reason for the edge.

Interpretation: Encodes pairwise convergence while distinguishing pathway co-membership, direct interaction, sequential dependency, and demonstrated interaction.

Status / limitation: Phase 5 model object; not inherently a causal or synergy matrix.

Project provenance: Phase 5

B.42 Directional consistency coding

$\text{direction} \in \{+1, 0, -1\}$

Variables / definitions: +1 = supports proposed mechanism; 0 = neutral/noninformative; -1 = contradicts.

Interpretation: Prevents counting any phenotype as supportive merely because a perturbation produces an effect.

Status / limitation: Phase 5 deterministic coding rule.

Project provenance: Phase 5

B.43 Ordinal evidence example

0=no relevant evidence; 1=weak/indirect; 2=moderate; 3=strong/direct; 4=replicated/direct/rescue-supported

Variables / definitions: Scale is illustrative unless formally frozen in the implementation.

Interpretation: Provides an interpretable ordinal mapping when continuous quantification is not justified.

Status / limitation: Phase 5 example scale; ordinal score is not a biological effect size or probability.

Project provenance: Phase 5

B.44 Empirical null-tail estimate (canonical)

$p_{\text{emp}} = (1 + \#\{T_{\text{null}} \geq T_{\text{index}}\}) / (N_{\text{null}} + 1)$

Variables / definitions: T_{index} = index-interval statistic; T_{null} = same statistic in matched null intervals; N_{null} = number of null intervals.

Interpretation: A standard way to quantify how extreme the index statistic is versus a simulated/matched null distribution.

Status / limitation: Canonical formula included for reader clarity because Phase 5 calls for empirical enrichment statistics but the accessible frozen protocol does not preserve a unique implemented formula.

Project provenance: Phase 5

B.45 Monte Carlo standard error for an estimated probability (canonical)

$$\text{MCSE}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/N}$$

Variables / definitions: $\hat{p}$ = simulated proportion; N = number of Monte Carlo replicates.

Interpretation: Shows why 20,000 replicates per scenario/cell can make stochastic sampling error small relative to biological uncertainty.

Status / limitation: Canonical simulation formula supplied for reader clarity; the Phase 5 summary reports replicate counts but not a stored MCSE equation.

Project provenance: Phase 5

Appendix C — Variable and Symbol Index

Symbol	Meaning in this project
A, B	Perturbation indicators or doses in the factorial interaction model.
ADP	Adenosine diphosphate.
ATP	Adenosine triphosphate.
B	Example Phase 5 bottleneck score.
C _i	Capacity of sequential process stage i.
C _i (Phase 5 locus vector)	Counterevidence/contradiction penalty for locus i; context distinguishes this from capacity C _i .
C _{max}	Available/maximal biological capacity in the functional-reserve condition.
C _s , C _e	Inclusive start/end coordinates of a CDS segment.
D	Damage generation in the layered-lesion equation; elsewhere “dosage consequence” in the Phase 5 causal chain.
D _i	Dosage evidence for locus i in the Phase 5 evidence vector.
D _s , D _e	Inclusive start/end coordinates of the deletion.
D _{early} , D _{late} , D _{adult}	Conceptual dosage effects in different developmental windows.
D(t)	Physiological demand as a function of time.
E _{i,k}	Evidence support contributed by locus i to mechanistic domain k.
ε	Residual/error term in the factorial interaction model.
F _C	Cortical/network transformation evidence in the Phase 5 causal chain.
F _D	Damage-response reserve domain.
F _E	Epileptogenic containment reserve domain.
F _k	Aggregated evidence for mechanistic domain k.
F _M	Membrane/endosomal handling reserve domain.
F _N	Neuronal developmental/network reserve; in the causal chain, neural/circuit consequence.
F _P	Photic phenotype relevance.
F _R	Retinal/photoreceptor reserve or retinal/visual-system consequence, depending on context.
f	Generic mapping function; also used as stimulus frequency when written G _R (f).
G	Genomic disruption in the Phase 5 causal chain; response gain in the Phase 4 gain definition, depending on context.
G _N	Network propagation gain.
G _R (f)	Retinal temporal gain at stimulus frequency f.
H	Photon/light exposure or dose.
H _{crit}	Exposure producing a predefined injury criterion.
I	Visual input in the photosensitive-network threshold; I(H) denotes measured injury as a function of exposure H in the light-injury model.
J _{in} , J _{out}	Input and output fluxes in the steady-state balance equation.
J _{max}	Approximate maximum throughput of a serial/bottleneck system.
K _{ij}	Pairwise convergence-matrix entry for loci i and j.
L	Nominal genomic interval length or lesion burden when written L(t), depending on context.
L _{CDS}	Total CDS length.
L _{deleted}	Total coding length overlapped/deleted.
L _{overlap}	Coding overlap length for one segment.
m	Number of hypothesis tests in BH-FDR notation.
M	Molecular/cellular perturbation in the causal chain.
M _i	Molecular perturbation evidence for locus i.
N _i	Neural/network relevance for locus i.
N _{null}	Number of null simulations/intervals in an empirical-tail estimate.
P	Prevention/tolerance in dL/dt = D – P – R; also mature phenotype in the developmental timing model; context is required.
P _i	Perturbation-equivalence evidence/weight for locus i.
$\hat{p}$	Estimated probability/proportion from Monte Carlo simulations.
Q	Target false discovery rate in BH procedure.
q	FDR-adjusted significance value.

Symbol	Meaning in this project
R	Repair capacity in the lesion-burden equation; $R(f)$ = retinal response amplitude at frequency f .
R_i	Rescue/causal-specificity evidence for locus i .
S	Spatiotemporal opportunity in the causal chain; $S(f)$ = stimulus amplitude at frequency f .
S_i	Spatiotemporal opportunity for locus i .
SD_g	Within-gene standard deviation across developmental stages.
T_{index}	Index-CNV test/convergence statistic in a null comparison.
T_{null}	Null-interval statistic generated under the same pipeline.
V_i	Visual-system relevance for locus i .
X	Generic molecular pool/concentration in the flux-balance equation.
Y	Outcome in the factorial interaction model.
β_0	Intercept in the factorial model.
β_A, β_B	Main-effect coefficients for perturbations A and B.
β_{AB}	Interaction coefficient; tests departure from the specified additive model.
Θ_E	Threshold for pathological synchronization in the conceptual photosensitive-network model.
ρ_s	Spearman rank-correlation coefficient.
Σ	Summation across indexed segments/loci.
Δ	Change in a measured quantity.

Appendix D — Interpretation and Reproducibility Rules

- Genomic deletion does not by itself establish RNA, protein, or functional dosage.
- Expression establishes biological opportunity, not dysfunction or causal effect.
- Knockout, biallelic loss, hypomorph, pathogenic missense/dominant-negative/gain-of-function states, and pharmacologic inhibition are not interchangeable with heterozygous deletion.
- Coexpression, pathway membership, network proximity, sequential dependence, biochemical interaction, genetic interaction, epistasis, and synergy are distinct evidence classes.
- An ordinal evidence score is not a biological effect size, probability, or percent causality.
- Missing evidence must remain missing. It must not be recoded as zero merely to complete a matrix.
- Zero should represent a defined null result or evidence of absence under stated conditions.
- Conceptual equations require independent calibration data before they can be interpreted quantitatively.
- Frequency scans and high-dimensional tests require prespecified correction for multiple comparisons.
- Cells from the same clone/donor are not independent biological replicates unless the experimental design explicitly supports that assumption.
- Null findings, contradictory effects, and failed hypotheses remain part of the frozen record.
- GRCh37/hg19 remains the controlling CNV coordinate system unless the project is formally versioned; GRCh38 annotations require documented conversion and cannot silently replace the fixed interval.
- The project's strongest current interpretation is a candidate distributed visual-neural functional-reserve lesion; direct causation of PPR or visually triggered seizures by the exact heterozygous CNV remains unresolved.

Appendix E — Provenance Note

This appendix was compiled from the frozen Phase 1 structural report, Phase 2 molecular synthesis, Phase 3 expression/spatiotemporal research records, Phase 4 frozen master report, the Phase 5 deterministic modeling protocol and frozen summary available in the project record, and the Phase 6 frozen conclusion. It does not reopen prior conclusions. Where a canonical formula was added because the frozen record named a standard method but did not print its equation (for example Spearman correlation, Benjamini-Hochberg correction, within-gene z-score, empirical null-tail probability, or Monte Carlo standard error), the entry is explicitly labeled as an explanatory mathematical definition rather than a recovered hidden calculation.